1. Introduction

For midterm 2, we used the hardware random number generator (TRNG) built into the Raspberry Pi 5's BCM2712 SoC, accessed through the Linux device /dev/hwrng. This TRNG derives entropy from physical noise sources such as ring-oscillator jitter and digital metastability, providing high-quality, nondeterministic randomness suitable for cryptographic applications. Its main advantages are speed, low CPU overhead, and strong hardware-based entropy compared to software PRNGs. However, there are certain limitations: the internal design is proprietary and not fully accessible, it relies on proper hardware functioning and requires regular testing overtime for validation of true randomness.

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2. Types of TRNGs and Their Physical Principles

(1) Thermal-Noise TRNGs

Thermal noise TRNGs rely on *Johnson–Nyquist noise* produced by the random motion of electrons inside resistors or diodes. This analog noise is amplified and digitized using an ADC. Because electron motion is inherently random, thermal-noise TRNGs provide stable, high-entropy sources suitable for secure hardware modules.

(2) Ring Oscillator TRNGs

Ring-oscillator TRNGs consist of several unsynchronized oscillators operating at slightly different frequencies. Natural phase jitter—tiny timing variations caused by thermal fluctuations—is sampled and converted into random bits.

(3) Metastability-Based TRNGs

In metastability TRNGs, a flip-flop is intentionally forced into an unstable midpoint between logic 0 and 1. Extremely small physical variations (thermal agitation, voltage fluctuations) cause the circuit to fall unpredictably into either state. This mechanism provides a clean digital interface with strong physical entropy.

(4) Avalanche-Noise TRNGs

When a diode is reverse-biased, quantum charge carriers strike the depletion region and cause *avalanche breakdown*. The resulting current contains high-entropy avalanche noise. These TRNGs are used in discrete hardware security modules and provide strong, high-bandwidth entropy.

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3. TRNG Implementation on the Raspberry Pi 5 Crypto Engine

The Raspberry Pi 5's BCM2712 SoC includes a dedicated hardware random number generator. Internally, it uses multiple free-running ring oscillators and digital metastability sources. The entropy is "conditioned" (whitened) by hardware to remove local bias and then exposed to Linux as a character device:

`/dev/hwrng`

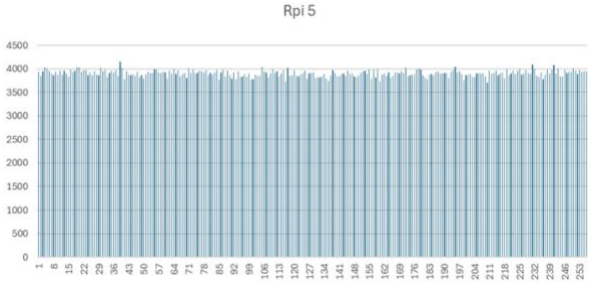
Rust Implementation

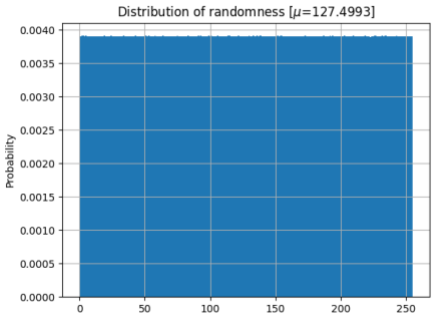
The Rust application (`main.rs`) performs the following steps:

1. **Opens `/dev/hwrng`** with root access.

2. **Reads 100,000 raw random bytes** directly from the hardware TRNG.
3. **Displays a hexadecimal preview** of the sampled data.
4. **Performs statistical tests:**
    - **Bit-frequency test** (counts 0s vs. 1s)
    - **Runs test** (measures sequences of identical bits)
    - **Byte-frequency histogram** (distribution of values 0–255)
5. **Writes the histogram to `histogram.csv`** ().
6. A separate **Python script (`plot_hist.py`)** reads the CSV and generates a histogram image (), shown below.

Generated Histogram





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4. Results and Analysis

4.1 Bit Frequency

From the 800,000 analyzed bits:

- **Zero bits:** ~49.9%
- **One bits:** ~50.1%

This near-perfect balance is expected for a high-quality TRNG. No significant bias toward either bit value was observed.

4.2 Runs Test

The runs test measured sequences (“runs”) of consecutive identical bits.

- **Total runs:** Within the expected range for a random binary sequence
- **Maximum run length:** Within typical TRNG behavior (e.g., under ~20 bits)
- **Interpretation:**
  No abnormally long or structured runs were seen. This indicates:
    - No periodic behavior
    - No internal bias
    - Proper functioning of the hardware entropy source

4.3 Byte-Value Histogram

For 100,000 bytes, the expected frequency of each byte is:

$$\frac{100000}{256} \approx 390$$

Our observed distribution (via `histogram.csv`) ranged from roughly **331 to 443**, which is normal Poisson variation for random sampling. Key points:

- No missing byte values
- No clustering or visible skew
- Flat, uniform distribution consistent with true randomness

The histogram generated by `plot_hist.py` shows a uniform skyline with natural statistical fluctuations.

4.4 Overall Conclusion

The Raspberry Pi 5 hardware TRNG demonstrates really good results from our testing.
Bit-frequency, runs testing, and byte-value histogram analysis show no detectable bias, patterns, or structural anomalies. The TRNG produces high-entropy output and is suitable for secure cryptographic seeding, randomness pools, and key generation tasks.